1 | Configuration and initialization

canopy_readnml biogenic, species, vertical, response and loss options

canopy_alloc allocate species arrays

canopy_init initialize history and emission arrays

2 | Driver preparation

Leaf light and temperature sunlit / shaded PPFD, Tleaf

Canopy and weather LAI, canopy height, temperature, soil, CO2, ozone, wind

hist_opt

0

0 | Use current PPFD, Tleaf, T2, wind

biospec_opt 0 = all 19 species 1...19 = selected species

1

1 | Update 24 h / 240 h means and daily stress extremes

leafage_opt=0 update LAI at lai_tstep

Biogenics enabled?

Yes

cluref > 0 (clumping)?

No

Skip biogenic module no biogenic output

No

Yes

3 | Species parameterization

CANOPY_BIO for each selected species

CANOPY_BIOP map species + vegetation to EF, LDF and response coefficients

Weather stress air quality, heat, cold, wind

Light response alpha, Cp from PPFD history sunlit + shaded leaves

Temperature response Topt, Eopt from Tleaf history sunlit + shaded leaves

Combined environment LDF x gamma_LDF + (1-LDF) x gamma_LIF

CO2 response isoprene only

Leaf-age response new to old foliage

Soil moisture response root-weighted layers

Layer activity EF x temperature/light x CCE x CO2 x leaf age x soil x stress responses

4 | Vertical treatment and conversion

biovert_opt

0

0 | Layer-resolved profile FLAI = delta FCLAI x LAI / dz emission = FLAI x activity

1

1 | Plant distribution VPGWT = FLAI / sum(FLAI)

2

2 | Gaussian weights five canopy levels

3

3 | Equal layer weights VPGWT = 1 / canopy layers

x 2.7777778e-13 microgram m-3 h-1 to kg m-3 s-1

Eligible target (LOSSIND = 0 or species)?

No

Yes, option 0

Yes, option 1

Yes, option 2

Loss off or ineligible CANLOSS = 1

Lifetime / wind / height canopy loss

Set loss factor CANLOSS = loss_set

Top-canopy flux LAI x weighted activity x CANLOSS

x 2.7777778e-13 microgram m-2 h-1 to kg m-2 s-1 store in top layer

Set selected profiles to zero

5 | Output

Point-list text 1D input; all species selected canopy_write_txt; *_bio.txt

Gridded NetCDF 2D input; NETCDF build canopy_write_ncf